Chen Siyu

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Job Intent: LLM Algorithm / LLM Application Development (Intern)

Education

Guangdong University of Technology Artificial Intelligence (Master's) **2024.09 – 2027.06**

Guangdong University of Technology Computer Science and Technology (Bachelor's) **2020.09 – 2024.06**

Publications & Patents

Text-Routed Mixture-of-Experts for Explainable Multimodal Sentiment Analysis **(AAAI-2026)** CCF-A, student first author

Explanation-Guided Multimodal Emotion Reasoning with Temporal Pre-Fusion **(ACM MM-2025 Challenge)** CCF-A, first author

Mamba-based Cross-Modal Temporal Fusion for Multimodal Large Language Models **(NeurIPS-2026, under review)** CCF-A, student first author

Research on Deep Learning-Based Aero-Engine Damage Recognition Technology (Peking University Core Journal 2024 + software copyright + patent) student first author

Research & Project Experience

Text-Routed Mixture-of-Experts Multimodal Sentiment Analysis Video sentiment analysis, temporal modeling, explainability **2024.11 – 2025.08**

Project Summary: To address the core problems in multimodal sentiment analysis that "text is dominant but under-utilized" and "audio-visual temporal information is hard to align", proposed a unified framework of "explanation-driven alignment + temporal modeling + text-routed mixture of experts", enhancing model interpretability while improving performance.

Core Work:

- **Explanation-driven Alignment:** Introduced an MLLM to generate emotional explanations of the audio and video modalities as an intermediate semantic bridge, mitigating cross-modal semantic inconsistency.
- **Temporal Alignment Modeling:** Designed Temporal Alignment to effectively model long- and short-term dependencies and synchronization between audio and video while keeping linear complexity.
- **Text-Routed MoE:** Proposed using text semantics as the routing signal to select expert networks, reinforcing the "text-dominant" property and reducing interference from ineffective modalities.
- **Experimental Results:** Outperformed existing methods on MOSI, MOSEI, CH-SIMS and other datasets; MAE on CH-SIMS dropped to 0.353 (↓13.5%).

Multimodal Emotion Reasoning and Explanation Generation Multimodal video understanding, emotion reasoning **2025.05 – 2025.09**

Project Summary: To address scarce annotation data in multimodal emotion recognition and the loss of temporal emotional information during LLM tokenization, proposed a unified "data augmentation + temporal pre-fusion" framework, achieving cross-modal emotion reasoning and natural-language explanation generation and significantly improving emotional consistency and interpretability.

Core Work:

- **Multimodal Emotion Reasoning Data:** Used AffectGPT fine-tuned on OV-MERD to generate reasoning-style emotion descriptions for SIMSv2; after Qwen3 consistency filtering and manual correction, built 4,403 high-quality emotion reasoning samples, effectively mitigating annotation scarcity and improving human-like reasoning ability.
- **Mamba Temporal pre-Fusion:** Before entering the LLM, performed cross-modal pre-fusion of audio and visual frame sequences, using SSM to model long-range temporal dependencies with linear complexity, then generated gating signals via Conv1d+SiLU to achieve bidirectional audio↔visual interaction through element-wise modulation; the fused features are projected into a unified multimodal representation that preserves temporal information, avoiding the loss of temporal dynamics and cross-modal synchronization caused by independent tokenization.
- **Multimodal Prompt Reasoning:** Built a structured Prompt combining fusion tokens, unimodal global tokens and subtitles, guiding Qwen2.5-7B to generate natural-language explanations with emotional reasoning chains. In the competition's official human evaluation, ranked **3rd** overall on the authenticity and fluency of the generated explanations.

CampGPT-X: LLM Training and Campus Intelligent Q&A LLM systems, intelligent Q&A **2025.12 – 2026.03**

Project Summary: Built a 2B-parameter decoder-only MoE large language model following the complete Pretraining → SFT → DPO pipeline, adapting a general language model to the campus Q&A domain; completed distributed training and inference optimization on a 4×A800 cluster and finally deployed a campus intelligent Q&A system.

Core Work:

- **Model Architecture & Pre-training:** Implemented a GPT-like architecture (RMSNorm, RoPE, SwiGLU, GQA) with 8-expert top-2 routing plus shared experts, mitigating expert collapse through load balancing; completed 100B-token pre-training, with validation PPL dropping from 38 to 17 and stable convergence.

- **Data Pipeline Construction:** Built domain data construction pipelines for the SFT/DPO stages, designing an automated "model generation + rule filtering" Pipeline that filters low-quality samples with relevance and consistency rules, reaching an 88% valid-sample ratio; finally built 10k instruction samples and 10k preference samples.
- **Instruction Tuning & Alignment:** Performed full-parameter SFT for domain adaptation, raising answer accuracy on the campus Q&A evaluation set from about 5% to 75%; further DPO preference optimization raised the pairwise win rate by about 15% and significantly reduced hallucinations.
- **Training & Inference Optimization:** Targeting memory and compute bottlenecks, used FSDP (ZeRO-2) for gradient and optimizer-state sharding during training, combined with Gradient Checkpointing and FlashAttention-2 to optimize memory and compute efficiency, reducing overall memory usage by 32% and increasing training throughput about 6×; at inference, reduced KV Cache overhead through GQA, cutting inference latency by about 40% and increasing throughput by about 60%.

Aero-Engine Damage Intelligent Detection and Deployment Industrial damage recognition **2024.02 – 2024.05**

Project Summary: Facing scarce aero-engine borescope image data and inefficient manual inspection, implemented damage recognition and terminal-device deployment based on YOLOv9.

Core Work: Responsible for borescope image annotation and data construction; expanded the sample size with a multi-region stitching Mixup method, then completed model fine-tuning and training-pipeline setup with transfer learning, finally achieving edge deployment (ONNX/TensorRT) to support inspection on embedded devices.

Professional Skills

- **LLM Fundamentals:** Familiar with PyTorch and Transformer, understand the principles and training workflows of mainstream LLMs, and master efficient fine-tuning methods such as PEFT.
- **Multimodal:** Familiar with VLM architectures such as CLIP, BLIP-2, LLaVA and Qwen-VL; master image-text alignment mechanisms and multimodal instruction tuning.
- **Reinforcement Learning:** Master RL fundamentals and can use mainstream alignment algorithms (PPO, DPO, GRPO, etc.) for model alignment and multi-objective optimization.